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Basics of Java Programming Lab Manual

For second Semester BE [VTU/CBCS, 2023-24 Syllabus]

Subject Code:

B	P	L	C	K	2	0	5	C



Name: …….

Branch: ……

Contents

1 Quadratic Equation 1

2 Array Multiplication 3

3 Shift Operation 5

4 Sorting 8

5 Student Details 10

6 Polymorphism - Method Overloading 14

7 Inheritance 16

8 Dynamic Dispatch 21

9 Packages and Acess modifiers 22

10 Exception Handling 27

Chapter 1

Quadratic Equation

Question

Write a JAVA program that prints all real solutions to the quadratic equation ax2+bx+c=0. Read in a, b, c and use the quadratic formula.

Java Code

```
import java.util.Scanner;

        public class QuadraticEquationDemo {
        public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print("Enter a: ");
        double a = input.nextDouble();

         System.out.print("Enter b: ");
         double b = input.nextDouble();

         System.out.print("Enter c: ");
         double c = input.nextDouble();

         double discriminant = b * b - 4 * a * c;
         if (discriminant > 0) {
                                System.out.println("The equation has real and distinct roots.");
                double root1 = (-b + Math.sqrt(discriminant)) / (2 * a);
                double root2 = (-b - Math.sqrt(discriminant)) / (2 * a);
                System.out.println("The roots are " + root1 + " and " + root2);
                } else if (discriminant == 0) {
                                System.out.println("The equation has real and equal roots.");
                double root = -b / (2 * a);
                System.out.println("The root is " + root);
                } else {
                                System.out.println("The equation has no real roots.");
                double realp = -b / (2 * a);
                double imagp = (Math.sqrt(-discriminant)) / (2 * a);
                System.out.printf("The roots are (%.2f + i%.4f) and (%.2f - i%.4f)\n", realp, imagp, realp, imagp);
                }
                }
```

```
}
```

Listing 1.1: QuadraticEquationDemo.java

Output

==========

```
putta:~/.../Program1$ javac QuadraticEquationDemo.java
putta:~/.../Program1$ java QuadraticEquationDemo
Enter a: 1
Enter b: 4
Enter c: 4
The equation has real and equal roots.
The root is -2.0

putta:~/.../Program1$ java QuadraticEquationDemo
Enter a: 1
Enter b: -5
Enter c: 6
The equation has real and distinct roots.
The roots are 3.0 and 2.0

putta:~/.../Program1$ java QuadraticEquationDemo
Enter a: 1
Enter b: 3
Enter c: 3
The equation has no real roots.
The roots are (-1.50 + i0.8660) and (-1.50 - i0.8660)
```

Listing 1.2: OUTPUT

Chapter 2

Array Multiplication

Question

Write a JAVA program for multiplication of two arrays.

Java Code

```
import java.util.Scanner;
import java.util.Arrays;

        public class MultiplyArraysDemo {
        public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
                        System.out.print("Enter the number of elements in the first array: ");
         int n = input.nextInt();
         int[] array1 = new int[n];
         for (int i = 0; i < n; i++) {
                                System.out.print("Enter element " + (i+1) + " of the first array: ");
                array1[i] = input.nextInt();
                }

                        System.out.print("Enter the number of elements in the second array: ");
         int m = input.nextInt();
         int[] array2 = new int[m];
         for (int i = 0; i < m; i++) {
                                System.out.print("Enter element " + (i+1) + " of the second array: ");
                array2[i] = input.nextInt();
                }

                if (n != m) {
       System.out.println("Error: Arrays have different length, not possible to multiply them.");
                  return;
           }

                int[] result = new int[n];
                for (int i = 0; i < n; i++) {
                result[i] = array1[i] * array2[i];
                }

      System.out.println("Result of multiplying arrays: " + Arrays.toString(result ));
      }
```

```
    }
```

Listing 2.1: MultiplyArraysDemo.java

Output

```
================================
putta:~/.../Program2$ javac MultiplyArraysDemo.java
putta:~/.../Program2$ java MultiplyArraysDemo
Enter the number of elements in the first array: 4
Enter element 1 of the first array: 1
Enter element 2 of the first array: 2
Enter element 3 of the first array: 3
Enter element 4 of the first array: 4
Enter the number of elements in the second array: 4
Enter element 1 of the second array: 4 Enter element 2 of the second array: 3
Enter element 3 of the second array: 2
Enter element 4 of the second array: 1
Result of multiplying arrays: [4, 6, 6, 4]

putta:~/.../Program2$ java MultiplyArraysDemo
Enter the number of elements in the first array: 3
Enter element 1 of the first array: 1
Enter element 2 of the first array: 2
Enter element 3 of the first array: 3
Enter the number of elements in the second array: 2
Enter element 1 of the second array: 4
Enter element 2 of the second array: 5
Error: Arrays have different length, not possible to multiply them.
```

Listing 2.2: OUTPUT

Chapter 3

Shift Operation

Question

Demonstrate the following operations and sign extension with Java programs i)

≪ ii) ≫ iii) ⋙

Java Code

```
import java.util.Scanner;

        class OperatorDemoSignExtension {
        public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("\nLeft Shift Operator\nEnter a number : ");
        int num = sc.nextInt();
     System.out.print("Enter the number of positions for the Left Shift Operation : ");
                int pos = sc.nextInt();

               int res = num << pos;
                        System.out.println("\nResult of Left Shift Operation");
                       System.out.printf("\n%d << %d = %d\n", num, pos, res);

         System.out.print("\nRight Shift Operator\nEnter a number : ");
         num = sc.nextInt();
    System.out.print("Enter the number of positions for the Right Shift Operation : ");
                pos = sc.nextInt();

              res = num >> pos;
                        System.out.println("\nResult of Right Shift Operation");
         System.out.printf("\n%d >> %d = %d\n", num, pos, res);
         if(num < 0){
         System.out.println("We observe that Even after the Right Shift operation the sign bit of negative number is
         preserved. We call this sign extension.");
         }
      System.out.print("\nUnsigned Right Shift Operator\nEnter a number : ");
      num = sc.nextInt();        System.out.print("Enter the number of positions for
the Unsigned Right Shift Operation : ");
                pos = sc.nextInt();

              res = num >>> pos;
```

```
        System.out.println("\nResult of Unsigned Right Shift Operation");
        System.out.printf("\n%d >>> %d = %d\n", num, pos, res);
            }}
```

Listing 3.1: OperatorDemoSignExtension.java

Output

```
putta:~/.../Program10$ javac OperatorDemoSignExtension.java
putta:~/.../Program10$ java OperatorDemoSignExtension

Left Shift Operator
Enter a number : 4
Enter the number of positions for the Left Shift Operation : 3

Result of Left Shift Operation

4 << 3 = 32

 Right Shift Operator
 Enter a number : -65
 Enter the number of positions for the Right Shift Operation : 2

Result of Right Shift Operation

 -65 >> 2 = -17
 We observe that Even after the Right Shift operation the sign bit of negative number is preserved. We call this sign
 extension

 Unsigned Right Shift Operator
 Enter a number : -65
 Enter the number of positions for the Unsigned Right Shift Operation : 2

Result of Unsigned Right Shift Operation

-65 >>> 2 = 1073741807

putta:~/.../Program10$ java OperatorDemoSignExtension

Left Shift Operator
Enter a number : 4
Enter the number of positions for the Left Shift Operation : 2

Result of Left Shift Operation

4 << 2 = 16

 Right Shift Operator
 Enter a number : 57
 Enter the number of positions for the Right Shift Operation : 2

Result of Right Shift Operation
```

```
57 >> 2 = 14

Unsigned Right Shift Operator
Enter a number : 57
Enter the number of positions for the Unsigned Right Shift Operation : 2

Result of Unsigned Right Shift Operation

57 >>> 2 = 14
```

Listing 3.2: OUTPUT

Chapter 4

Sorting

Question

Write a JAVA program to sort list of elements in ascending and descending order.

Java Code

```
import java.util.Scanner;

        public class ExchangeSortDemo {
        public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
                        System.out.println("Enter the number of elements in the array: ");
        int n = sc.nextInt();
        int[] arr = new int[n];

                        System.out.println("Enter the elements of the array: ");
    for (int i = 0; i < n; i++) { arr[i] = sc.nextInt();
          }

                            for (int i = 0; i < n - 1; i++) {
                            for (int j = i + 1; j < n; j++) {
                            if (arr[i] > arr[j]) {
                            int temp = arr[i];
                            arr[i] = arr[j];
                            arr[j] = temp;
                            }
                            }
                            }

                        System.out.println("\nSorted array in ascending order: ");
                for (int i = 0; i < n; i++) {
                System.out.print(arr[i] + " ");
                }

                            for (int i = 0; i < n - 1; i++) {
                            for (int j = i + 1; j < n; j++) {
                            if (arr[i] < arr[j]) {
                            int temp = arr[i];
                            arr[i] = arr[j];
                            arr[j] = temp;
```

```
                    }
                    }
                    }

        System.out.println("\nSorted array in descending order: "); CHAPTER 4. SORTING


                    for (int i = 0; i < n; i++) {
                    System.out.print(arr[i] + " ");
                    }

    }
    }
```

Listing 4.1: ExchangeSortDemo.java

Output

```
================================
putta:~/.../Program4$ javac ExchangeSortDemo.java
putta:~/.../Program4$ java ExchangeSortDemo Enter the number of elements in the array:
6
Enter the elements of the array:
6 1 5 2 4 3

Sorted array in ascending order:
1 2 3 4 5 6
Sorted array in descending order:
6 5 4 3 2 1

putta:~/.../Program4$ java ExchangeSortDemo Enter the number of
elements in the array:
5
Enter the elements of the array:
9 1 8 2 5

Sorted array in ascending order:
1 2 5 8 9
Sorted array in descending order:
9 8 5 2 1
```

Listing 4.2: OUTPUT

Chapter 5

Student Details

Question

Create a JAVA class called Student with the following details as variables within it. USN
NAME
BRANCH
PHONE
PERCENTAGE
Write a JAVA program to create n Student objects and print the USN, Name, Branch, Phone, and percentage of these objects with suitable headings.

Java Code

```
import java.util.Scanner;
import java.util.List;
import java.util.ArrayList;

        class StudentType {
        private String USN;
        private String NAME;
        private String BRANCH;
        private String PHONE;
                private double PERCENTAGE;

                  public StudentType(String USN, String NAME, String BRANCH, String PHONE, double PERCENTAGE) {
                  this.USN = USN;
                  this.NAME = NAME;
                  this.BRANCH = BRANCH;
                  this.PHONE = PHONE;
                  this.PERCENTAGE = PERCENTAGE;
                  }

                  public String getUSN() {
                     return USN;
          }

                 public String getNAME() {
                    return NAME;
          }

         public    String    getBRANCH()     {
return BRANCH;
```

```
30          }
31
32          public String getPHONE() { 33
            return PHONE;
34                  }
35
36          public    double   getPERCENTAGE()          {
            37 return PERCENTAGE;
38          }
39          }
40
41
42                  public class StudentDemo {
43                  public static void main(String[] args) {
44                  Scanner sc = new Scanner(System.in);
45                  List<StudentType> students = new ArrayList<>();
46
47          System.out.println("Enter the number of students:"); 48          int n = sc.nextInt();
49
50                          for (int i = 1; i <= n; i++) {
51                                  System.out.println("Enter the details of student " + i + ":");
52                      System.out.print("USN: ");
53                      String USN = sc.next();
54                      System.out.print("NAME: ");
55                      String NAME = sc.next();
56                      System.out.print("BRANCH: ");
57                      String BRANCH = sc.next();
58                      System.out.print("PHONE: ");
59                      String PHONE = sc.next();
60                      System.out.print("PERCENTAGE: ");
61                      double PERCENTAGE = sc.nextDouble();
62
63                      students.add(new StudentType(USN, NAME, BRANCH, PHONE, PERCENTAGE));
64                      }
65                      System.out.println("\nSTUDENT DETAILS\n====================");  66      System.out.println("USN" +
                        "\t\t" +
"NAME" + "\t" + "BRANCH" + "\t" + "PHONE " + "\t\t" + "PERCENTAGE");
67                  for (StudentType student : students) {
68                  System.out.println(student.getUSN() + "\t" + student.getNAME() + "\t" + student.getBRANCH() + "\t" +
                    student.getPHONE() + "\t" + student.getPERCENTAGE()) ;
69                  }
70                  }
71                  }
```

Listing 5.1: StudentDemo.java

Output

==========

```
1 putta:~/.../Program5$ javac StudentDemo.java
```

```
putta:~/.../Program5$ java StudentDemo Enter the number of students:
2
Enter the details of student 1:
USN: 1SI22CS036
NAME: Rajesh
BRANCH: CSE
PHONE: 7187238165
PERCENTAGE: 89.7

Enter the details of student 2:
USN: 1PE22EC088
NAME: Sheela BRANCH: ECE
PHONE: 8238887233
PERCENTAGE: 84.5

STUDENT DETAILS
====================
USN NAME      BRANCH PHONE PERCENTAGE
1SI22CS036 Rajesh CSE 7187238165 89.7
1PE22EC088 Sheela ECE 8238887233 84.5
```

Listing 5.2: OUTPUT

Chapter 6

Polymorphism - Method Overloading

Question

Write a JAVA program demonstrating Method overloading and Constructor overloading.

Java Code

```
class PointType {
private double x;        private double y;

        // Constructor overloading
        public PointType() {
        this.x = 0;   this.y = 0;
        }

            public PointType(double x, double y) {
         this.x = x;
         this.y = y;
         }

                    // Method overloading to calculate distance between two PointType objects
             public double distance(PointType point) {
                                          System.out.println("\nCalculating distance between two PointType objects");
         return Math.sqrt(Math.pow(x - point.x, 2) + Math.pow(y - point.y, 2));
         }

    // Method overloading to calculate distance between a PointType object and a point specified by its coordinates
              public double distance(double x, double y) {
     System.out.println("Calculating distance between a PointType object and a point specified by its coordinates");
     return Math.sqrt(Math.pow(this.x - x, 2) + Math.pow(this.y - y, 2));
          }

        public void show(){
          System.out.printf("(%.1g,%.1g)\n",x,y);
          }
          }

          class PointDemo{
          public static void main(String[] args) {
          PointType point1 = new PointType(1, 1);
          PointType point2 = new PointType(7, 9);
          System.out.print("\npoint1 coordinates :"); point1.show();
```

```
38              System.out.print("point2 coordinates :"); point2.show();
39              double dVal = point1.distance(point2);
40                              System.out.println("Distance between point1 and point2: " + dVal );
41
42              PointType point3 = new PointType();
43              System.out.print("\npoint3 coordinates :"); point3.show();
44              dVal = point3.distance(3,4);
45                              System.out.println("Distance between point3 and (3, 4): " + dVal);
46      }
47      }
```

Listing 6.1: PointDemo.java

Output

```
==========
1  putta:~/.../Program6$ javac PointDemo.java
2  putta:~/.../Program6$ java PointDemo
3
4  point1 coordinates :(1,1)
5  point2 coordinates :(7,9)
6
7  Calculating distance between two PointType objects
8  Distance between point1 and point2: 10.0
9
10 point3 coordinates :(0,0)
11 Calculating distance between a PointType object and a point specified by its coordinates 12 Distance
   between point3 and (3, 4): 5.0
```

Listing 6.2: OUTPUT

Chapter 7

Inheritance

Question

Design a super class called Staff with details as StaffId, Name, Phone, Salary. Extend this class by writing three subclasses namely Teaching (domain, publications), Technical (skills), and Contract (period). Write a JAVA program to read and display at least 3 staff objects of all three categories.

Java Code

```
class Staff {
                private int staffId;
        private String name;
        private String phone;
                private double salary;

                public Staff(int staffId, String name, String phone, double salary) {
                this.staffId = staffId;
                this.name = name;
                this.phone = phone;         this.salary = salary;
        }
        public int getStaffId() { return staffId;
        }

                public String getName() {
                   return name;
        }

                public String getPhone() {
                return phone;
                }

        public   double  getSalary()       {
return salary;
        }

                public void DisplayInfo(){
                System.out.println("Name: " + this.getName());
                System.out.println("Staff ID: " + this.getStaffId());
                System.out.println("Phone: " + this.getPhone());
                System.out.println("Salary: " + this.getSalary());
                }
```

```
class Teaching extends Staff {
private String domain;
private int publications;

public Teaching(int staffId, String name, String phone, double salary, String domain, int publications) {
super(staffId, name, phone, salary);
this.domain = domain;
this.publications = publications;
} 
public String getDomain() {         return domain;
 }

public int getPublications() {         return publications;
 }

        public void DisplayInfo(){
        super.DisplayInfo();
        System.out.println("Domain: " + this.getDomain());
        System.out.println("Publications: " + this.getPublications());
        }
        }

class Technical extends Staff {    private   String skills;

public Technical(int staffId, String name, String phone, double salary, String skills) {
super(staffId, name, phone, salary);          this.skills = skills;
 }

public String getSkills() { return skills;
        }
        public void DisplayInfo(){
        super.DisplayInfo();
        System.out.println("Skills: " + this.getSkills());
        }
        }

class Contract extends Staff {    private   int period;

public Contract(int staffId, String name, String phone, double salary, int period) {
super(staffId, name, phone, salary);          this.period = period;
 }

public int getPeriod() { return period;
        }
        public void DisplayInfo(){
        super.DisplayInfo();
        System.out.println("Period: " + this.getPeriod()+" months");
        }
        }
```

```
97
98
99          public class StaffTypeDemo {
100         public static void main(String[] args) {
101         Teaching t1 = new Teaching(1, "Rajesh Nayak", "9822546534", 75000, "Computer
        Science", 15);
102         Teaching t2 = new Teaching(2, "Sita Devi", "8787432499", 80000, "Mathematics ", 20);
103         Teaching t3 = new Teaching(3, "John Peter", "8528734373", 85000, "Physics",
        25);
104         Technical te1 = new Technical(4,"Ramesha","9473673642",90000,"Java,Python,C ++"); 105 Technical te2 = new
        Technical(5,"Suresha","8917612332",95000,"JavaScript, React,Node.js"); 106 Technical te3 = new
        Technical(6,"Dinesha","9944222323",100000,"Python, TensorFlow,Keras");
107         Contract c1 = new Contract(7, "Abida Begum", "9323786211", 75000, 6);
108         Contract c2 = new Contract(8, "Lily Thomas", "8776551219", 80000, 12);
109         Contract c3 = new Contract(9, "Seema Jain", "9922324343", 85000, 18);
110             //display the staff objects
111         System.out.println("Teaching Staff 1:");
112         t1.DisplayInfo();
113         System.out.println("\nTeaching Staff 2:");
114         t2.DisplayInfo();
115         System.out.println("\nTeaching Staff 3:"); 116 t3.DisplayInfo();
117
118     System.out.println("\nTechnical Staff 1:"); 119         te1.DisplayInfo();
120
121     System.out.println("\nTechnical Staff 2:"); 122         te2.DisplayInfo();
123
124     System.out.println("\nTechnical Staff 3:"); 125         te3.DisplayInfo();
126
127     System.out.println("\nContract Staff 1:"); 128          c1.DisplayInfo();
129
130     System.out.println("\nContract Staff 2:"); 131          c2.DisplayInfo();
132
133     System.out.println("\nContract Staff 3:"); 134          c3.DisplayInfo();
135      }
136      }
```

Listing 7.1: StaffTypeDemo.java

Output

```
================================
1  putta:~/.../Program7$ javac StaffTypeDemo.java
2  putta:~/.../Program7$ java StaffTypeDemo 3 Teaching Staff 1:
4  Name: Rajesh Nayak
5  Staff ID: 1
6  Phone: 9822546534
7  Salary: 75000.0
8  Domain: Computer Science
9  Publications: 15
10
11 Teaching Staff 2:
12 Name: Sita Devi
```

Staff ID: 2
Phone: 8787432499
Salary: 80000.0
Domain: Mathematics
Publications: 20

Teaching Staff 3:
Name: John Peter
Staff ID: 3
Phone: 8528734373
Salary: 85000.0
Domain: Physics
Publications: 25

Technical Staff 1:
Name: Ramesha
Staff ID: 4
Phone: 9473673642
Salary: 90000.0
Skills: Java,Python,C++

Technical Staff 2:
Name: Suresha
Staff ID: 5
Phone: 8917612332
Salary: 95000.0
Skills: JavaScript,React,Node.js

Technical Staff 3:
Name: Dinesha
Staff ID: 6
Phone: 9944222323
Salary: 100000.0
Skills: Python,TensorFlow,Keras

Contract Staff 1:
Name: Abida Begum
Staff ID: 7
Phone: 9323786211
Salary: 75000.0
Period: 6 months

Contract Staff 2:
Name: Lily Thomas
Staff ID: 8
Phone: 8776551219
Salary: 80000.0
Period: 12 months

Contract Staff 3:
Name: Seema Jain
Staff ID: 9

Phone: 9922324343
Salary: 85000.0
Period: 18 months

Listing 7.2: OUTPUT

Chapter 8

Dynamic Dispatch

Question

Demonstrate dynamic dispatch using abstract class in JAVA.

Java Code

```
abstract class Shape {    abstract void draw();
}

        class Circle extends Shape {
        void draw() {
        System.out.println("Drawing Circle");
        }
        }

         class Square extends Shape {
         void draw() {
         System.out.println("Drawing Square");
         }
         }

         class DynamicDispatchDemo {
         public static void main(String[] args) {
         Shape s;
         s = new Circle();
         s.draw();
         s = new Square();
         s.draw();
         }
         }
```

Listing 8.1: DynamicDispatchDemo.java

Output

```
putta:~/.../Program8$ javac DynamicDispatchDemo.java
putta:~/.../Program8$ java DynamicDispatchDemo
Drawing Circle
Drawing Square
```

Listing 8.2: OUTPUT

Chapter 9

Packages and Acess modifiers

Question

Create two packages P1 and P2. In package P1, create class A, class B inherited from A, class C . In package P2, create class D inherited from class A in package P1 and class E. Demonstrate working of access modifiers (private, public, protected, default) in all these classes using JAVA.

Java Code

```
1 // Package P1
2 package com.P1;
3
4 // Class A with private and protected variables and public methods
5 public class A { 6 private int x; 7 protected int y;
8
9 public void setX(int x) { 10 this.x
= x;
11 }
12
13 public int getX() { 14 return x;
15 }
16
17 public void setY(int y) { 18 this.y
= y;
19 }
20
21 public int getY() { 22 return y;
23 }
24 }
```

Listing 9.1: A.java

```
1 // Package P1
2 package com.P1;
3
4
5 // Class B inherited from A with default variable and public method
6 public class B extends A { 7 int z;
8
9 public void setZ(int z) {
10 this.z = z;
11 }
```

```
12

13      public int getZ() { 14          return z;
15      }
16      }
```

Listing 9.2: B.java

```
1 // Package P1
2 package com.P1;
3
4 // Class C with private variable and public method
5 public class C { 6 private int w;
7
8       public void setW(int w) { 9
        this.w = w;
10              }
11
12      public int getW() { 13          return w;
14      }
15      }
```

Listing 9.3: C.java

```
1 // Package P2
2 package com.P2;
3
4 // Class D inherited from A with protected variable and public method
5 public class D extends com.P1.A { 6
protected int v;
7
8       public void setV(int v) { 9
        this.v = v;
10              }
11
12      public int getV() { 13          return v;
14      }
15      }
```

Listing 9.4: D.java

```
1 // Package P2
2 package com.P2;
3
4 // Class E with private variable and public method
5 public class E { 6 private int u;
7
8       public void setU(int u) { 9
        this.u = u;
10              }
11
```

```
    public int getU() {        return u;
    }
```

Listing 9.5: E.java

```
import com.P1.A;
import com.P1.B;
import com.P1.C;
import com.P2.D;
import com.P2.E;

public class PackageDemo {

    public static void main(String[] args) {

        // Creating an object of class A in package P1
        A a = new A();
        a.setX(10);
        a.setY(20);
        System.out.println("Value of x: " + a.getX());
        System.out.println("Value of y: " + a.getY());

        // Creating an object of class B in package P1
        B b = new B();
        b.setX(30);
        b.setY(40);
        b.setZ(50);
        System.out.println("Value of x: " + b.getX());
        System.out.println("Value of y: " + b.getY());
        System.out.println("Value of z: " + b.getZ());

        // Creating an object of class C in package P1
        C c = new C();
        c.setW(60);
        System.out.println("Value of w: " + c.getW());

        // Creating an object of class D in package P2
        D d = new D();
        d.setX(70);        d.setY(80);
        d.setV(90);
        System.out.println("Value of x: " + d.getX());
        System.out.println("Value of y: " + d.getY());
        System.out.println("Value of v: " + d.getV());
```

```
        // Creating an object of class E in package P2
        E e = new E();
        e.setU(100);
        System.out.println("Value of u: " + e.getU());
    }
}
```

Listing 9.6: PackageDemo.java

The above files should be stored in the folder structure as shown in the below diagram.

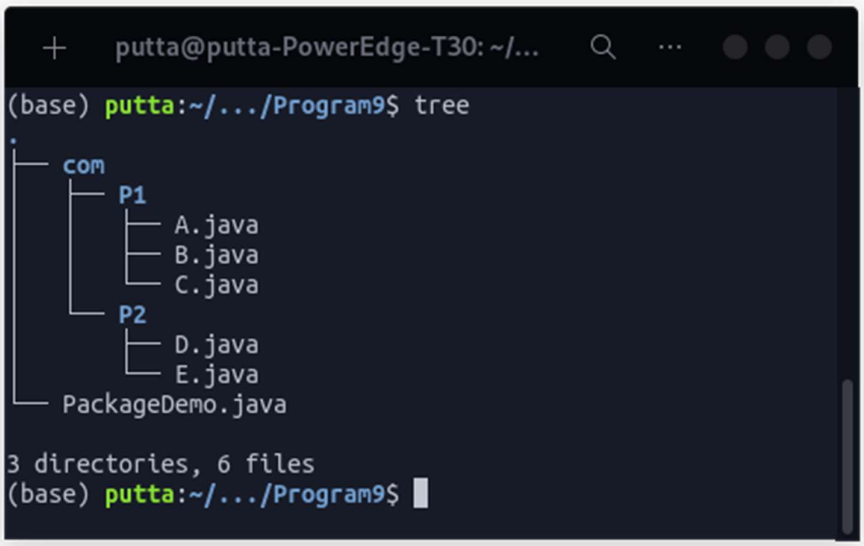


Figure 9.1: Package Structure

First create a folder **com** in the same folder where you saved *PackageDemo.java*. Then within **com** directory create two directories **P1** and **P2**. Within **P1** directory store the files *A.java*, *B.java* and *C.java*. Within **P2** directory store the files *D.java* and *E.java*.

Output

```
==================================
putta:~/.../Program9$ javac PackageDemo.java
putta:~/.../Program9$ java PackageDemo
Value of x: 10    Value of y: 20    Value of x: 30    Value of y: 40
Value of z: 50
Value of w: 60
Value of x: 70
Value of y: 80
Value of v: 90
Value of u: 100
```

Listing 9.7: OUTPUT

Chapter 10

Exception Handling

Question

Write a JAVA program to read two integers a and b. Compute a/b and print, when b is not zero. Raise an exception when b is equal to zero. Also demonstrate working of ArrayIndexOutOfBoundException.

Java Code

```
import java.util.Scanner;

        public class ExceptionDemo {
        public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter two integers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
           try {
                    if (b == 0) {
                    throw new ArithmeticException("Division by zero");
                    }
                    int result = a / b;
                    System.out.println("Result of integer division: " + result);
                    } catch (ArithmeticException e) {
                    System.out.println(e.getMessage());
                    }
                    try {
                    int[] arr = new int[5];
                   arr[10] = 50;                    //invalid index generates exception
                } catch (ArrayIndexOutOfBoundsException e) {
                System.out.println(e.getMessage());
                }
                }
                }
```

Listing 10.1: ExceptionDemo.java

Output

```
==================================
putta:~/.../Program10$ javac ExceptionDemo.java
putta:~/.../Program10$ java ExceptionDemo

```

```
Enter two integers:
9 2
Result of integer division: 4
Index 10 out of bounds for length 5

putta:~/.../Program10$ java ExceptionDemo

Enter two integers:
3 0
Division by zero
Index 10 out of bounds for length 5
```

Listing 10.2: OUTPUT24